BCS304 – DATA STRUCTURES AND APPLICATIONS

MODULE WISE QUESTION BANK

(THEORY AND PROBLEM QUESTIONS)

THEORY QUESTIONS

MODULE – 1: Introduction, Stacks & Strings

Topic: Data Structure Fundamentals

- Define Data Structures and explain the classification with a neat diagram and examples. **(MQP, July 24, Supply July 24, Jan 25, July 25)**
- Explain all operations of Data Structures **(**Create, Insert, Delete, etc.**).** **(Jan 24, July 24, Supply July 24)**
- Differentiate between Primitive and Non-Primitive data structures. **(MQP, Jan25)**

Topic: Memory & C Programming

- Explain dynamic memory allocation functions **(**malloc, calloc, realloc, free**)** in detail with syntax and examples. **(MQP, July 24, Jan 25, July 25, Makeup July 25)**
- Differentiate between structures and unions with examples. **(Jan 24, Jan 25, Makeup July 25)**
- Discuss self-referential structures with a suitable example. **(July 24)**

Topic: Stacks & Applications

- Define stack and implement push**()**, pop**()**, and display**()** operations using arrays. **(MQP, Jan 24, July 24, Jan 25, July 25, Makeup July 25)**
- Define stack. With suitable example write the array representation of stack. **(July 25)**
- Explain ADT stack. **(July 24)**

Topic: Pattern Matching & Strings

- What do you mean by pattern matching? Outline the Knuth-Morris-Pratt **(**KMP**)** pattern matching algorithm with an example. **(MQP, Jan 24, Makeup July 25)**
- Write C functions for: i) Compare strings, ii) Concatenate strings, iii) Reverse strings without built-in functions. **(MQP)**

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- Explain any five string handling functions supported by 'c' with syntax and example. **(Supply July 24)**

Topic: Sparse matrix

- What is Sparse matrix? **(Makeup July 25)**

MODULE – 2: Queues & Linked Lists

Topic: Linear & Circular Queues

- Develop a C program to implement insertion, deletion and display operations on Linear queue. **(MQP)**
- Discuss the implementation of circular queue operations **(**insert, delete, display**)** and explain its advantages over linear queues. **(MQP**, **Jan 24**, **July 24**, **Supply July 24**, **Jan 25**, **July 25**, **Makeup July 25)**
- With suitable example discuss the representation of linear queue with array. **(July 2025)**
- What are the disadvantages of linear queue? **(MQP)**
- What are the disadvantages of ordinary queue? **(Jan 24)**
- Define Queue and discuss how to represent it using dynamic arrays. **(Jan 24**, **Jan 25)**

Topic: Multiple Stacks & Queues

- Write a note on multiple stacks and multiple queues with suitable diagrams and examples. **(July 24**, **Jan 24**, **Jan 25**, **Makeup July 25)**
- Explain in detail about multiple queues with relevant functions in 'C'. **(Supply July 24)**
- Write a note on priority queue. **(Jan 24)**
- Explain Linked Stacks and Queues operations. **(July 24)**

Topic: Singly Linked Lists (SLL)

- What is a linked list? Explain the different types with neat diagrams. **(Jan 24**, **Jan 25**, **July 25)**

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- Write C functions for SLL:
  i) Insert node at beginning/end
  ii) Delete node at front/end
  iii) Display.
  iv) Concatenation of two SLL
  v) Inverting Singly linked list
  (Jan 24, July 24, Jan 25, July 25, Makeup July 25)

- Write a program in C to implement a stack of integers using a singly linked list. **(MQP, July 25)**

- Develop a function to delete a node whose information field is specified in singly linked list. **(July 24)**

- Write C functions for:
  i) Searching an element
  ii) Concatenating two SLLs
  iii) Inverting an SLL. **(MQP, Jan 25, July 25)**

- Develop a linked list with the basic operations performed on Singly Linked List (SLL) and different types oflinked list. **(Supply July 24)**

- Examine a node structure for linked representation of polynomial. **(Supply July 24)**

Topic: Polynomials

- Write a C function to add two polynomials using SLL/Circular SLL. **(MQP, Jan 24, Supply July 24, Jan 25, Makeup July 25)**

MODULE – 3: Advanced Lists & Trees

Topic: Doubly & Circular Linked Lists

- Define Doubly Linked List (DLL) and write structure definitions. **(July 25)**

- Write a structure definition for Doubly Linked List (DLL) of integers. What are the advantages of DLL over SLL? **(July 25)**

- Write C functions for DLL:
  i) Inserting a node at front/rear
  ii) Deleting a node at front/rear
  iii) Search a node with a given key value
  iv) Concatenation of two DLL

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v) Finding the length of a circular list.
(Jan 24, Supply July 24, Jan 25, July 25, Makeup July 25)

- Implement Circular DLL operations:
  i) Insert at beginning
  ii) Delete at end.
  iii) Display
  (July 24)

Topic: Binary Tree Basics

- Define Binary tree**(**BT**)**. Explain the representation of a binary tree with a suitable example. **(MQP)**
- Define Tree and explain terminology **(**Complete BT, Height, Degree, Level, Depth, Path, Full BT**)**. [Learn minimun 6 terminologies]
  (July 24, Supply July 24, Makeup July 25)
- Discuss how binary trees are represented using: i) Array, ii**)** Linked list.
  (Jan 24, July 24, Jan 25, July 25)
- Write function for copying and testing of binary tree. **(Supply July 24)**

Topic: Tree Traversals

- Write recursive C functions for Inorder, Preorder, and Postorder traversals of a binary tree. **(MQP, Jan 24, Supply July 24, Jan 25, July 25, Makeup July 25)**
- Discuss Level order traversal with a recursive function. **(Jan 24, Jan 25)**

Topic: Threaded Binary Tree

- Define Threaded Binary Tree and discuss In-Threaded binary tree.
  (MQP, July 24, Jan 24, Jan 25)
- What are the advantages of threaded binary tree over a binary tree? Construct Threaded Binary Tree for given elements **(**A , B, C, D, E, F, G, H, I**)**.
  (MQP, Jan 25, July 25, Makeup July 25)

MODULE – 4: BST & Graphs

Topic: Binary Search Tree (BST)

- Define BST and list its common operations. **(July 24, Jan 25)**

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- Write a function to perform recursive search and insertion in a BST. **(MQP, Jan 24, Supply July 24, July 25, Makeup July 25)**

Topic: Forests & Selection Trees

- Define Forest and explain the procedure to transform a forest into a binary tree with an example. **(MQP, Jan 24, July 24, Jan 25, July 25)**
- Define selection tree and explain types **(**Winner/Loser**)** with examples. **(MQP, Jan 24, Jan 25, July 25)**

Topic: Graphs

- Define Graph and explain Adjacency Matrix and Adjacency List representation with an example. **(MQP, Jan 24, Supply July 24, Jan 25, July 25)**
- Define terminology: Digraph, Weighted graph, Self loop, Parallel edges, Connected graph. **(Jan 24, Jan 25)**
- Explain in detail the elementary graph operations. **(July 24, Jan 24, Jan 25)**
- Define graph. Explain graph abstract data types. **(July 24)**
- Write a C function to traverse a graph using Depth First Search **(**DFS**)** and Breadth First Search **(**BFS**). (MQP, Supply July 24, July 25, Makeup July 25)**
- Compare a graph with a tree.

Topic: Disjoint Sets

- Build a linked list representation of disjoint sets. **(Supply July 24)**
- Explain simple find and collapsing find. **(MQP)**

MODULE – 5: Hashing & Heaps

Topic: Hashing

- Define hashing and explain different hashing functions **(**Division, Mid-square, etc.**)** with examples. Discuss the properties of a good hash function. **(MQP, July 24, Jan 24, Jan 25, July 25, Makeup July 25, Supply July 24)**
- Explain Static and Dynamic hashing in detail with operations. **(MQP, July 24, Jan 24, Supply July 24, Jan 25, July 25, Makeup July 25)**
- What is chained hashing? Discuss pros and cons. **(MQP)**

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- What is collision? Explain methods to resolve collision **(**Linear probing**)** with algorithm/example. **(Jan 24**, **Supply July 24** , **Jan 25**, **Makeup July 25)**

Topic: Priority Queues & Heaps

- Explain Priority Queue with example? Demonstrate Max Priority Queue operations **(**Insert, Delete, Display**). (MQP**, **July 24**, **Jan 25**, **July 25)**
- Describe double ended priority queue. **(Supply July 24)**

Topic: Advanced Trees

- Define/Explain Leftist Trees **(**HBLT/WBLT**)** with examples. Give its declaration in C. **(MQP**, **July 24**, **Jan 24**, **Jan 25**, **July 25**, **Makeup July 25)**
- Write a short note on Optimal Binary Search Tree **(**OBST**).** **(MQP**, **Jan 24**, **July 24**, **Jan 25**, **July 25)**
- Discuss AVL Tree, Red-Black Tree, and Splay tree with insertion methods. **(Jan 24)**

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PROBLEMS

MODULE – 1: Stacks, Expressions & KMP

Topic: Postfix Evaluation

- Write a function to evaluate the postfix expression. Illustrate the same for the given postfix expression: ABC-D*+ESF+ assuming A=6, B=3, C=2, D=5, E=1, F=7. **(MQP)**
- Write an algorithm/function to evaluate a postfix expression and apply the same for the given postfix expression. 62/3-42*+

  (MQP, Jan 24, Jan 25, Makeup July 25)
- Write a C function to evaluate postfix expression. Evaluate 231*+9- using the same. **(Makeup July 2025)**

Topic: Infix to Postfix Conversion

- Write the Postfix form using stack:

  (i) A*(B*C+D*E)+F **(ii)** (a+(b*c)/(d-e)) **(Jan 2024)**
- Convert the following infix expression to postfix using stack:

  A+(B*C-(D/E^F)*G)*H. **(Supply July 2024)**
- Translate the following infix expressions to postfix form:

  (i) ((A*B)+C)/D **(ii)** A*B*C+D **(July 2025)**

Topic: Pattern Matching (KMP Algorithm)

- Illustrate the KMP algorithm to find the occurrences of pattern P: ABCDABD in text S: ABC ABCDAB ABCDABCDABDE. **(MQP)**
- Write KMP pattern matching algorithm and search the pattern "abcdabcy" in the text "abcxabedabxabcdabcdabey". **(Makeup July 2025)**

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Topic: Sparse Matrix (Triplet Form & Transpose)

- Give the triplet form for the given matrix and find its transpose: **(July 2024)**

A =

	0	1	2	3
0	10	0	0	40
1	11	0	22	0
2	0	0	0	0
3	20	0	0	50
4	0	15	0	25

- Give the triplet form of the given matrix and find its transpose: **(Jan 2025)**

$$A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$$

- Draw the triplet representation and the transpose of resultant triplet: **(July 2025)**

$$A = \begin{bmatrix} 15 & 0 & 0 & 22 & 0 & -15 \\ 0 & 11 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 91 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 28 & 0 & 0 & 0 \end{bmatrix}$$

MODULE – 2: Queues & Polynomials

Topic: Polynomial Addition (Linked Representation)

- Write the C function to add two polynomials. Show the linked representation of the below two polynomials and their addition using a circular singly linked list
  P1: 5x3 + 4x2 +7x + 3
  P2: 6x2 + 5
  Output: add the above two polynomials and represent them using the linked list. **(MQP)**

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- Write a C-function to add two polynomials show the linked representation of below two polynomials:

$$p(x) = 3x^{14} + 2x^{8} + 1$$
$$q(x) = 8x^{14} - 3x^{10} + 10x^{6}$$

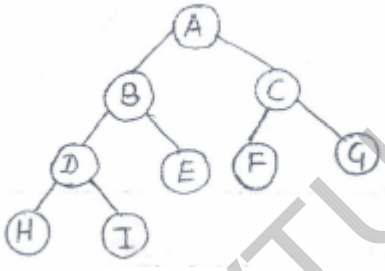
(Jan 2024)

- Examine a node structure and explain algorithm to add two polynomials. **(Supply July 2024)**
- Write a C function to add two polynomials. **(Jan 2025)**
- Develop a C function to add two polynomials using Single Linked List. **(Makeup July 2025)**

MODULE – 3: Linked Representations & Traversals

Topic: Binary Tree

- Write array and linked list representation for given binary tree.



(July 2025)

- Design threaded binary tree for the given elements: 10,20,30,40,5 **(July 2025)**

Topic: Sparse Matrix (Linked List Representation)

- For the given sparse matrix, give the diagrammatic linked representation for:

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 1 \\ 0 & 0 & 6 & 0 \end{bmatrix}.$$

(Jan 2024)

VTUADDA

- For the given sparse matrix, give the linked list representation:

$$A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$$

(MQP)

- Summarize Sparse Matrix. For the given sparse matrix, write the diagrammatic linked list representation for:

$$\begin{bmatrix} 8 & 0 & 0 & 0 \\ 5 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 8 \\ 0 & 0 & 9 & 1 \end{bmatrix}$$

(Supply July 2024)

- For the given sparse matrix, design the linked list representation for:

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 1 \\ 0 & 0 & 6 & 0 \end{bmatrix}$$

(July 2025)

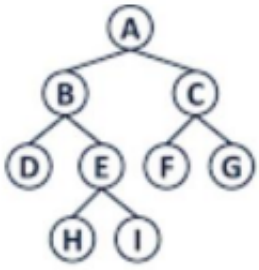
- Give the diagrammatic representation of below sparse matrix using Linked list:

$$\begin{bmatrix} 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 10 \\ 3 & 1 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(Makeup July 2025)

Topic: Tree Construction & Traversals

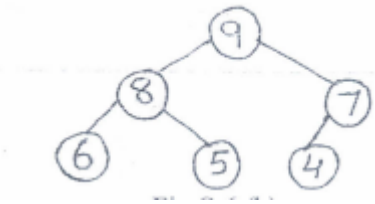
- Find all traversals for the given tree.



(MQP)

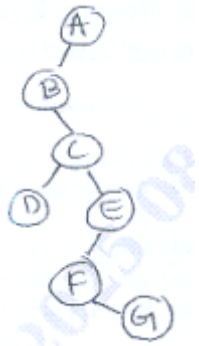
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- Draw a binary tree and find out the binary tree traversals for the expression: 3+4*(7-6)/4+3. **(Supply July 2024)**
- Find inorder, preorder, and postorder traversals for the given binary tree.



(July 2025)

- Traverse given tree using Inorder, Preorder, and Postorder traversal.



(Makeup July 2025)

- For a given data design a binary search tree. Apply inorder, preorder and postorder traversals on resultant binary search tree. 6,9,5,2,8,15,**24**, 14,7,8,5,2 **(July 2025)**

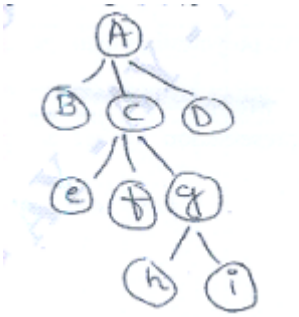
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MODULE – 4: BST, Winners Trees & Forests

Topic: BST Construction

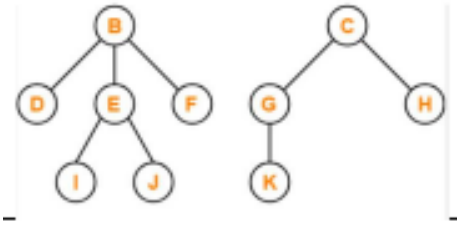
- Construct a binary search tree **(**BST**)** for the following elements: 100, 85, 45, 55, 120, 20, 70, 90, 115, 65, 130, 145. Traverse using in-order, pre-order, and post-order traversal techniques. Write recursive C functions for the same. **(MQP)**
- Draw a BST for: 100, 85, 45, 55, 110, 20, 70, 65. **(July 2024)**
- Construct BST and perform traversals for: 14, 15, 4, 9, 7, 18, 3, 5, 16, 20. **(Supply July 2024)**
- Construct a BST for: 100, 85, 45, 55, 120, 20, 70, 90, 115, 65, 130, 145. **(Jan 2025)**
- Design a BST and apply traversals for: 6, 9, 5, 2, 8, 15, **24**, 14, 7, 8, 5, 2. **(July 2025)**

- Convert the following forest into Binary Tree and perform inorder, preorder and postorder traversals.



(Makeup July 2025)

- Transform the given forest into a Binary tree and traverse using inorder, preorder and postorder traversal.



(MQP)

- Construct a binary tree from sequences:
  In-order: GDHBAEICF
  Post-order: GHDBIEFCA.
  (MQP)

- Construct min winner tree for the runs of a game given below. Each run consists of values of players. Find the first 5 winners:

10	9	20	6	8	9	90	17
15	20	20	15	15	11	95	18
16	38	30	25	50	16	99	20
			28				

(MQP)

- Construct first 4 winners from runs given below:

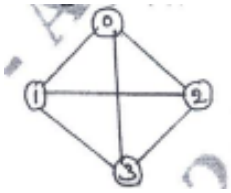
5	9	18	8	90	09	7	8
14	13	19	11	95	17	13	10
24	16	20	15	98	19	14	15

(Makeup July 2025)

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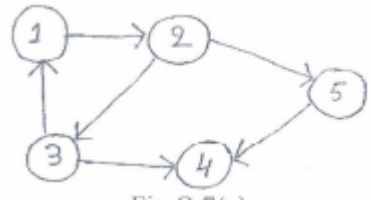
Topic: Adjacency matrix and Adjacency linked list

- Show the adjacency matrix and adjacency list representation of the graph given below



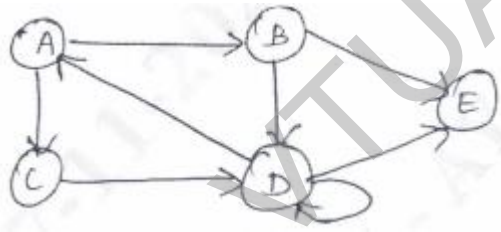
(Jan 2024)

- Write a adjacency matrix and adjacency linked list representation for following given graph



(July 2025)

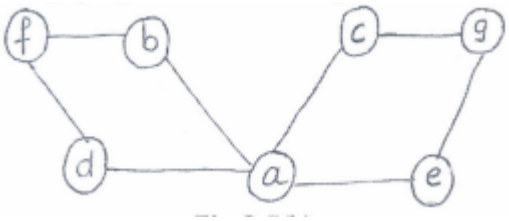
- For the graph shown below show the adjacency matrix and adjacency list representation.



(Supply July 2024)

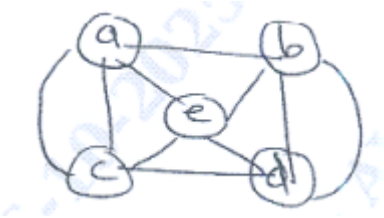
Topic: BFS & DFS

- Design an algorithm to traverse a graph using Depth First Search **(**DFS**)**. Apply DFS for the graph given below starting from C.



(MQP, July 2025)

- Design the alogorithm to traverse below graph using BFS and DFS technique. Traverse the below graph using BFS from source vertex.



(Makeup July 2025)

- Consider the tree created by the weighted union function on the sequence of unions: union**(**0,1**)**, union**(**2,3**)**, union**(**4,5**)**, union**(**6,7**)**, union**(**0,2**)**, union**(**4,6**)**, and union**(**0,4**)**. Process the simple find and collapsing find on eight finds and compare which find is efficient. **(MQP)**

MODULE – 5: Hashing & OBST

Topic: Collision Resolution (Linear Probing)

- Insert keys 72, 27, 36, 24, 63, 81, 92, 101 into % [size 10] table using linear probing. **(Supply July 2024)**
- Explain linear probing for the data 12, 13, 15, 16, 74, 75, 83, 42. **(Makeup July 2025)**

Topic: Chained Hashing

- Construct the hash table to insert keys: 7, 24,18, 52, 36, 54, 11, 23 in a chained hash table of 9 memory locations using h**(**k**)** = k mod m. **(MQP)**

Topic: Optimal Binary Search Tree (OBST)

- Construct an OBST for the following keys with the probabilities as

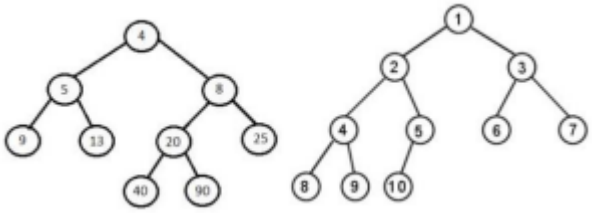
Keys	A	B	C	D	E
Probability	0.25	0.2	0.05	0.2	0.3

(Supply July 2024)

- Construct the OBST for following items. Let n=4
  (a1, a2, a3, a4 **)** = **(** do, if, int, while **)**
  Let **(** p1, p2, p3, p4 **)** = **(** 3, 3, 1, 1 **)**
  (q0, q1, q2, q3, q4 **)**=**(** 2, 3, 1, 1, 1 **)**. **(Makeup July 2025)**

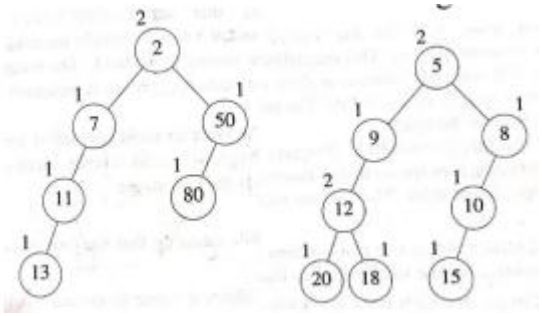
Topic: Leftist tree

- Check whether the given binary tree is a leftist tree or not. Explain your answer.



(MQP)

- Meld the given min leftist trees.



(MQP)

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